AWS Fundamental

Today's Lesson

Cloud computing

AWS Regions and Availability Zones

AWS Basic service: EC2, VPC

Cloud computing

What is cloud computing?

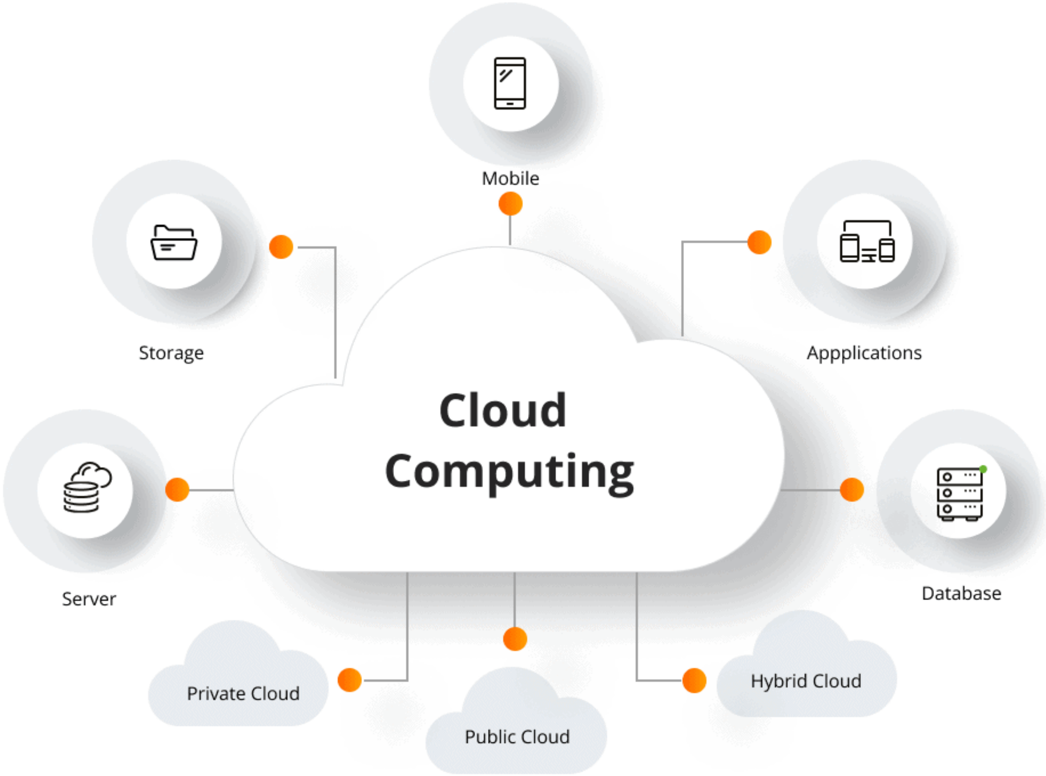
Cloud computing is the delivery of computing services—such as servers, storage, databases, networking, software, and more—over the internet ("the cloud") instead of on local computers or physical hardware.

Key idea

You access computing resources remotely, on demand, paying only for what you use. This eliminates the need to own or manage physical infrastructure directly.

Deployment models

- Public cloud – Resources are provided by third-party vendors over the internet.
- Private cloud – Used exclusively by one organization.
- Hybrid cloud – Combines public and private resources for flexibility.
- Multi-cloud – Uses multiple cloud providers to avoid dependency on a single one.

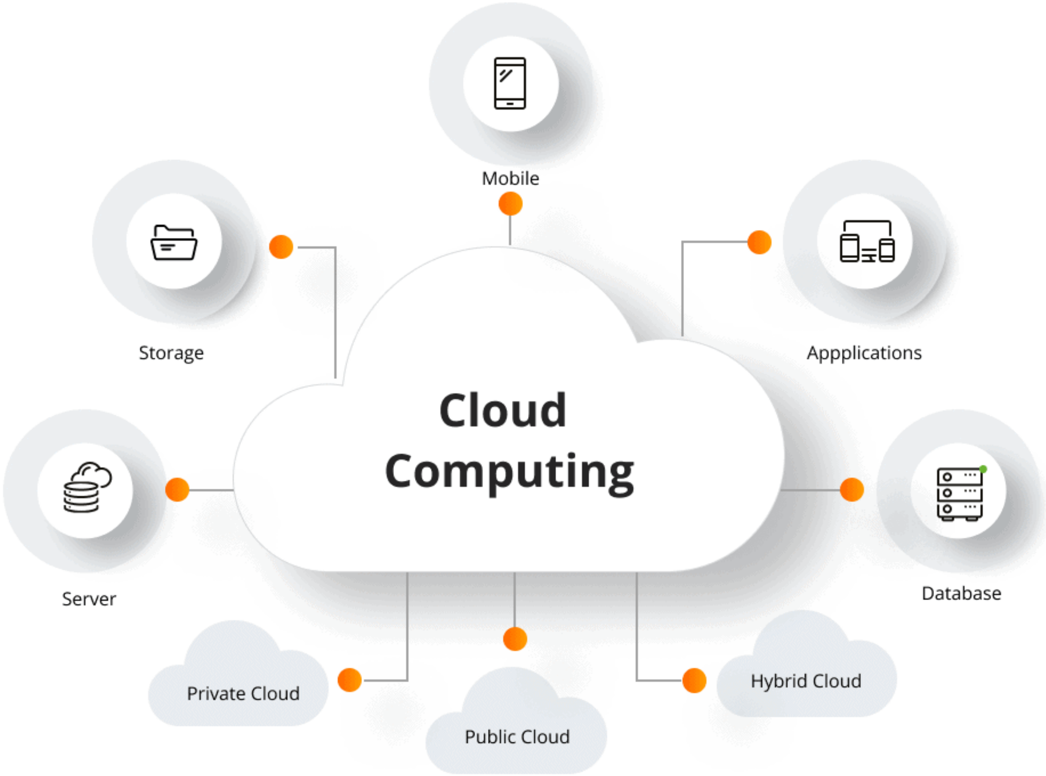


Service Model

Infrastructure as a Service (IaaS) – Provides virtual machines, storage, and networks (e.g., Amazon EC2, Google Compute Engine).

Platform as a Service (PaaS) – Offers a platform for building, testing, and deploying applications (e.g., Google App Engine, Microsoft Azure App Service).

Software as a Service (SaaS) – Delivers ready-to-use applications over the internet (e.g., Gmail, Salesforce).



Public Cloud market share

As of **2024**, the **public cloud market** is dominated by a few major providers often referred to as "**hyperscalers**." These companies offer infrastructure, platform, and software services to millions of customers globally.

Rank	Provider	Parent Company	Estimated Global Market Share
1	Amazon Web Services (AWS)	Amazon	~31–32%
2	Microsoft Azure	Microsoft	~25–26%
3	Google Cloud Platform (GCP)	Alphabet (Google)	~11%
4	Alibaba Cloud	Alibaba Group	~4–5%
5	Oracle Cloud Infrastructure (OCI)	Oracle Corporation	~2–3%
6	IBM Cloud	IBM	~2%
7	Tencent Cloud	Tencent	~2%
–	Others (DigitalOcean, Huawei Cloud, VMware, etc.)	–	~18–20% (combined)

Amazon web service

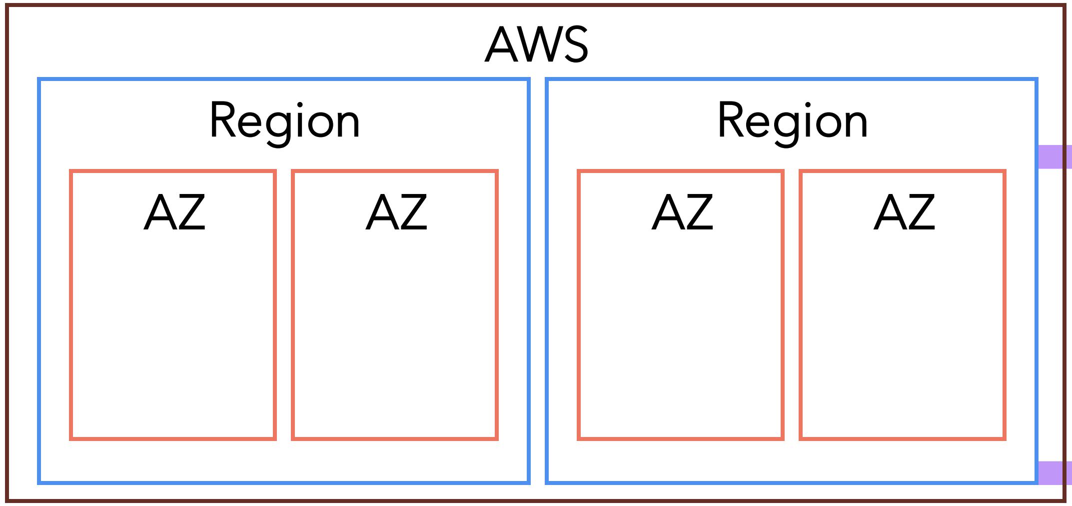
AWS Region and Zone

Region

- An **AWS Region** is a **geographically distinct area** where AWS has a group of data centers.
  Each region is **independent**, with its own *power, networking, and connectivity*—this ensures **data sovereignty** and **fault isolation** from other regions.

AWS Availability Zone (AZ)

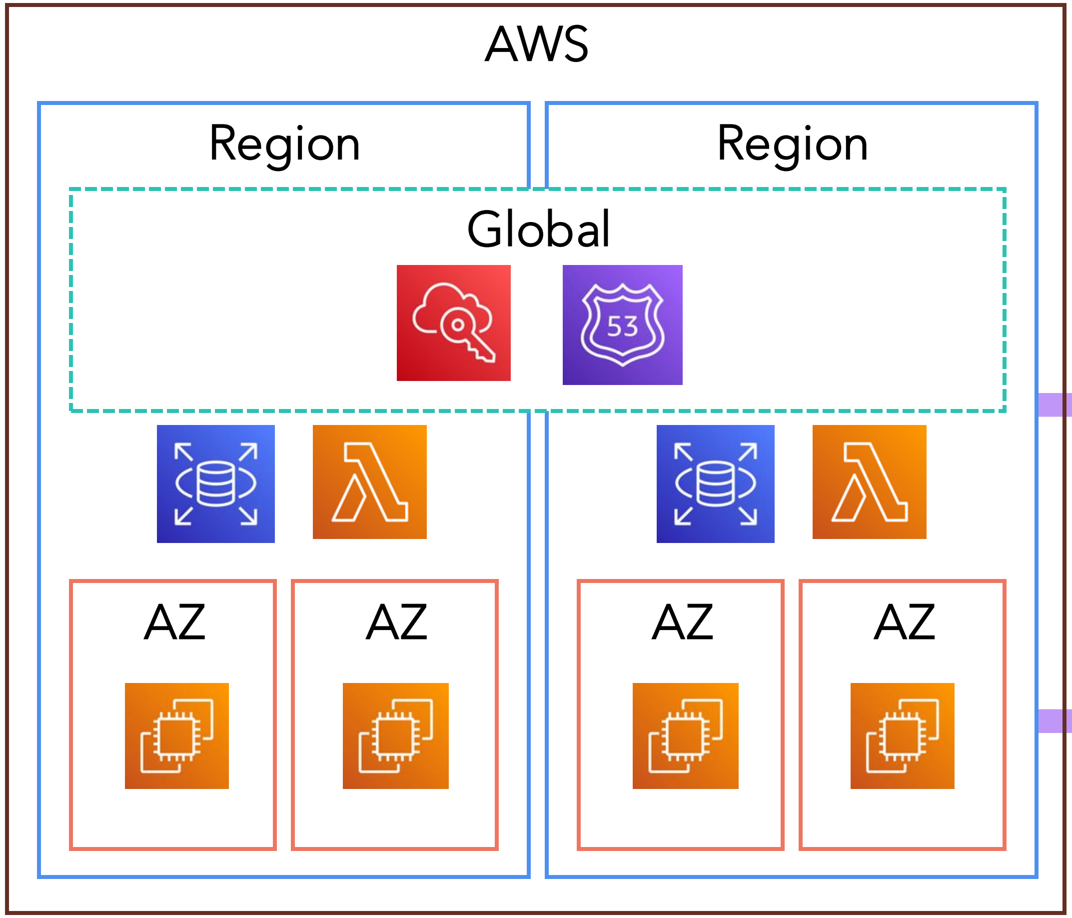
- An **Availability Zone (AZ)** is one or more **physically separate data centers** within a Region.
  Each AZ has **independent power, cooling, and networking**, connected by **low-latency, high-speed links**.



AWS Region and Zone

Service type:

- Global
    - Operates **across all AWS regions worldwide**.
    - Amazon Route 53, CloudFront (CDN), IAM, AWS WAF, AWS Organizations
- Regional
    - Operates **across multiple AZs in one region** – automatically replicated or managed for fault tolerance
    - S3, RDS (Multi-AZ), ECS, AWS Lambda, VPC,…
- Zonal
    - Operates **within a single Availability Zone (AZ)** – a physically separate data center inside a region.
    - Amazon EC2 Instance, EBS Volume, Elastic Load Balancer (in a single AZ), …



AWS Region and Zone

This is an example of Region and AZ supported by AWS

Region Name	Region Code	Location	Example Availability Zones
US East (N. Virginia)	us-east-1	United States	us-east-1a, us-east-1b, us-east-1c...
US West (Oregon)	us-west-2	United States	us-west-2a, us-west-2b,...
Europe (Ireland)	eu-west-1	Ireland	eu-west-1a, eu-west-1b,...
Asia Pacific (Tokyo)	ap-northeast-1	Japan	ap-northeast-1a, ap-northeast-1b,...
Asia Pacific (Mumbai)	ap-south-1	India	ap-south-1a, ap-south-1b,...
South America (São Paulo)	sa-east-1	Brazil	sa-east-1a, sa-east-1b, sa-east-1c
Middle East (Bahrain)	me-south-1	Bahrain	me-south-1a, me-south-1b, me-south-1c
Africa (Cape Town)	af-south-1	South Africa	af-south-1a, af-south-1b, af-south-1c

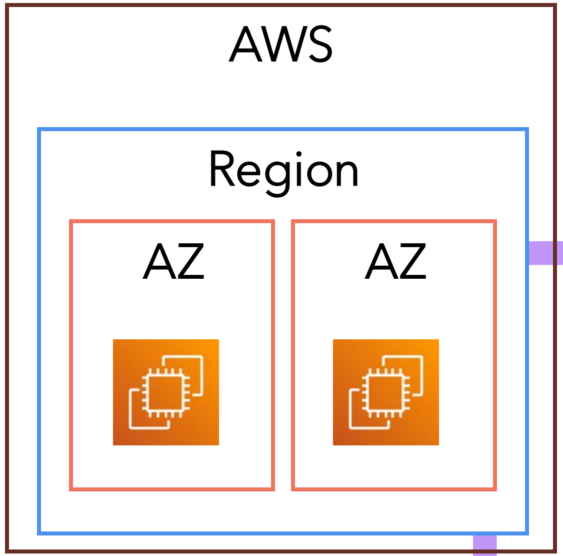
AW

AZ

Elastic Compute Cloud (EC2)

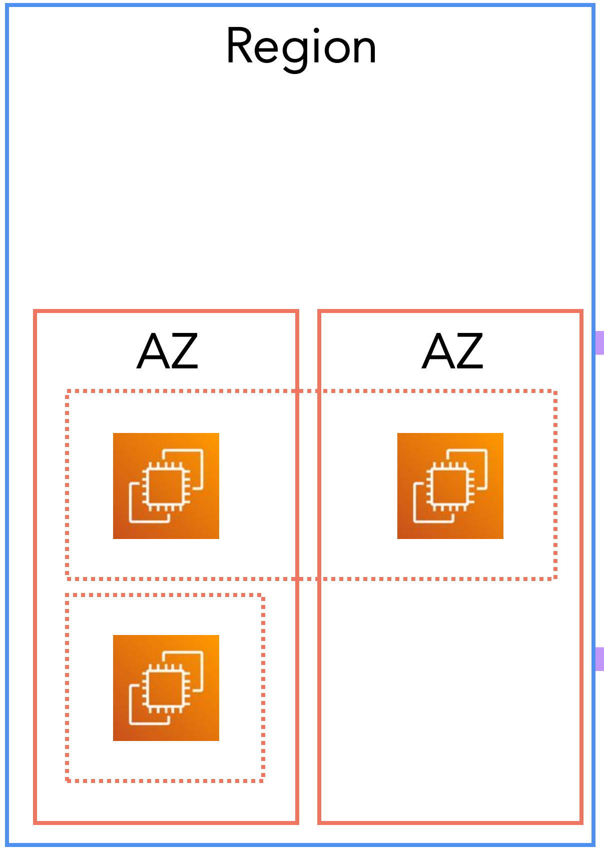
- **Amazon EC2 (Elastic Compute Cloud)** is one of the core services in AWS – it provides **resizable, on-demand virtual servers** in the cloud that you can use to run applications, host websites, process data, and more.

Term	Meaning
Instance	A virtual machine running on the AWS cloud.
AMI (Amazon Machine Image)	A pre-configured template containing an OS and optional software to launch an instance.
Instance Type	Defines the hardware resources (vCPU, memory, storage, network) for your instance (e.g., t3.micro, m6i.large).
EBS (Elastic Block Store)	Persistent storage volumes attached to EC2 instances.
Elastic IP	A static, public IPv4 address you can assign to an EC2 instance.
Key Pair	Used to securely connect to your instance via SSH (Linux) or RDP (Windows).
Security Group	Virtual firewall controlling inbound and outbound traffic.



Security Group

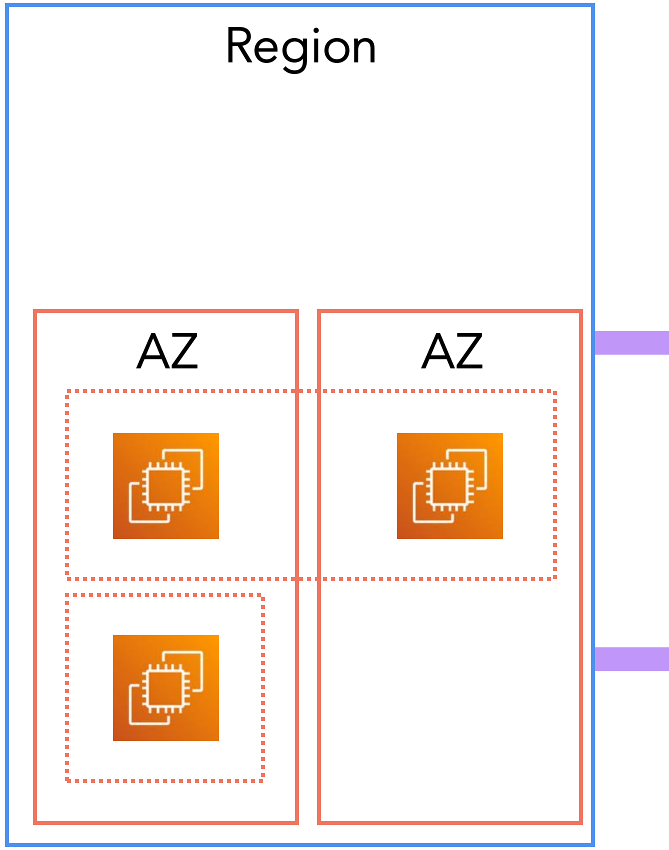
- A **Security Group (SG)** in AWS is a **virtual firewall** that **controls inbound and outbound network traffic** for your **Amazon EC2 instances** (and some other AWS resources).
- It determines **which traffic is allowed** to enter or leave your instance at the **instance level**, not at the subnet or network level.
- Each EC2 instance is associated with **one or more Security Groups**.
- When you **launch an instance**, you specify one or more SGs.
- Each rule in a Security Group defines:
  - **Protocol** (e.g., TCP, UDP, ICMP)
  - **Port range** (e.g., 22 for SSH, 80 for HTTP)
  - **Source/Destination** (e.g., IP address, CIDR block, or another SG)



Security Group

- Example: Typical Web Server Setup

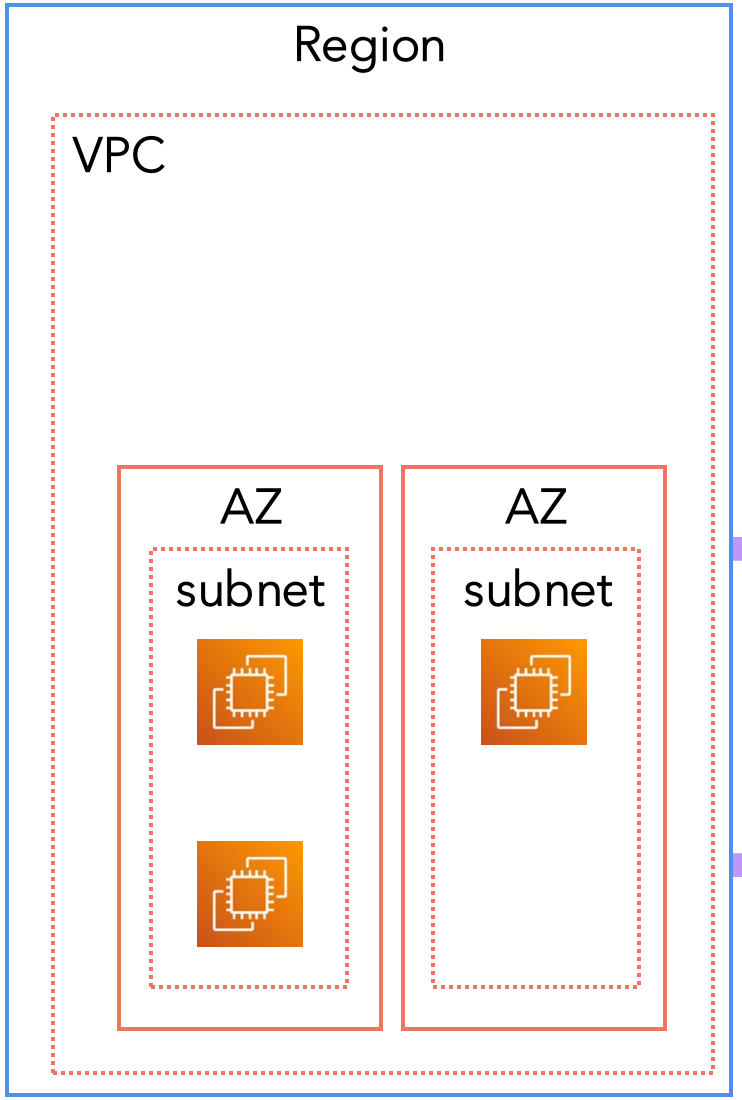
Rule Type	Protocol	Port	Source/Destination	Purpose
Inbound	TCP	22	Your IP address (203.x.x.x/32)	Allow SSH for admin access
Inbound	TCP	80	0.0.0.0/0	Allow HTTP traffic from anywhere
Inbound	TCP	443	0.0.0.0/0	Allow HTTPS traffic from anywhere
Outbound	All	All	0.0.0.0/0	Allow all outbound traffic



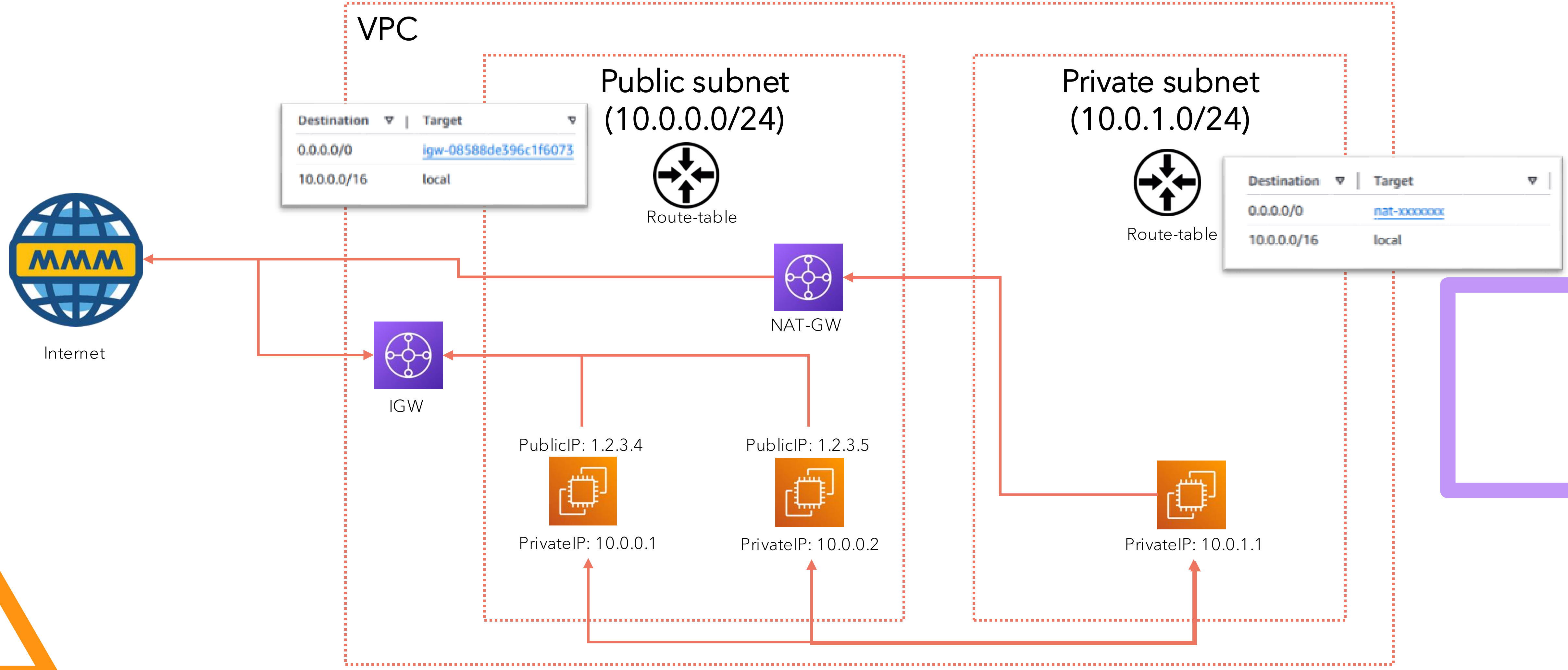
Virtual Private Cloud (VPC)

- **A VPC is your own private section of the AWS cloud**, with complete control over IP addressing, routing, security, and connectivity.
- Common Component

Component	Role
VPC	The overall virtual network environment.
Subnet	Logical divisions inside the VPC; each subnet is tied to an **Availability Zone** (AZ).
Internet Gateway (IGW)	Enables public internet access to/from VPC resources (e.g., web servers).
NAT Gateway	Allows private instances to access the internet securely (for software updates, etc.).
Route Table	Defines traffic rules – decides where packets go (internet, other subnets, VPN, etc.).



Virtual Private Cloud (VPC)



Subnet

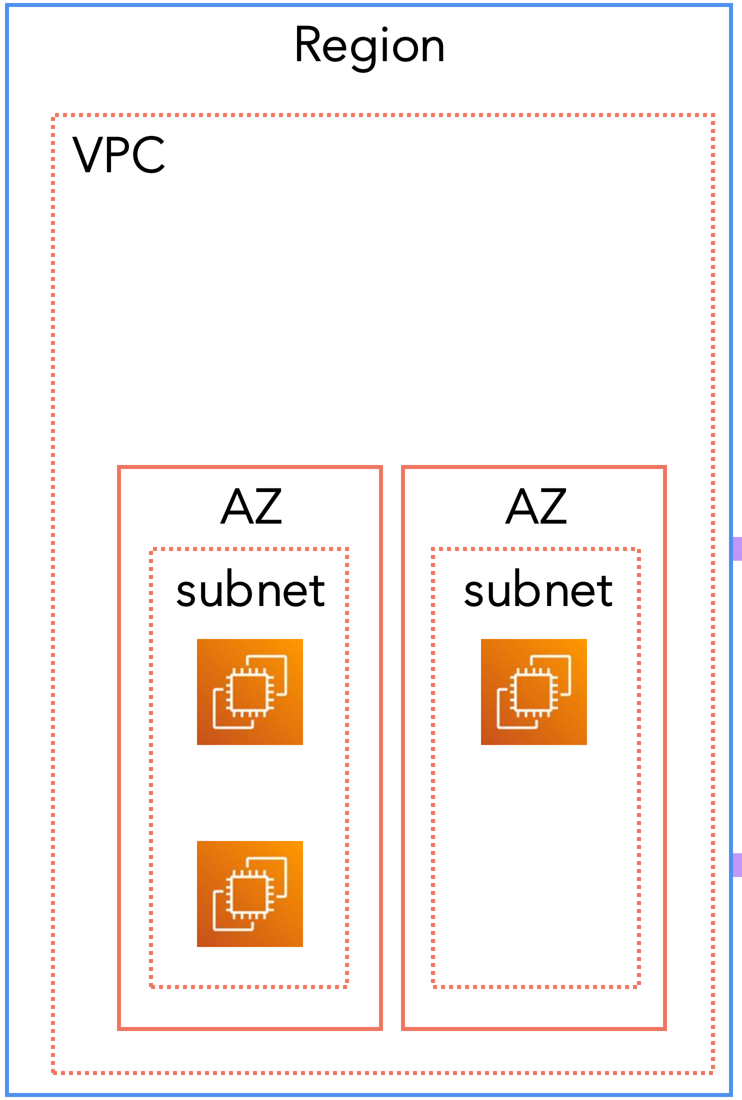
A **Subnet** is a logical subdivision of an AWS **VPC** (Virtual Private Cloud). Each subnet must reside entirely within *one* **Availability Zone (AZ)**.

You use subnets to separate resources based on access needs – for example, web servers (public) and databases (private).

Example:

- VPC CIDR block: **10.0.0.0/16**
- Subnet 1 (Public): **10.0.1.0/24**
- Subnet 2 (Private): **10.0.2.0/24**

Type	Description	Example Use
Public Subnet	Connected to the internet through an **Internet Gateway (IGW)**	Hosts web servers that need public access
Private Subnet	Has no direct internet access; may connect via **NAT Gateway** for outbound traffic	Hosts application servers or databases



Internet Gateway (IGW)

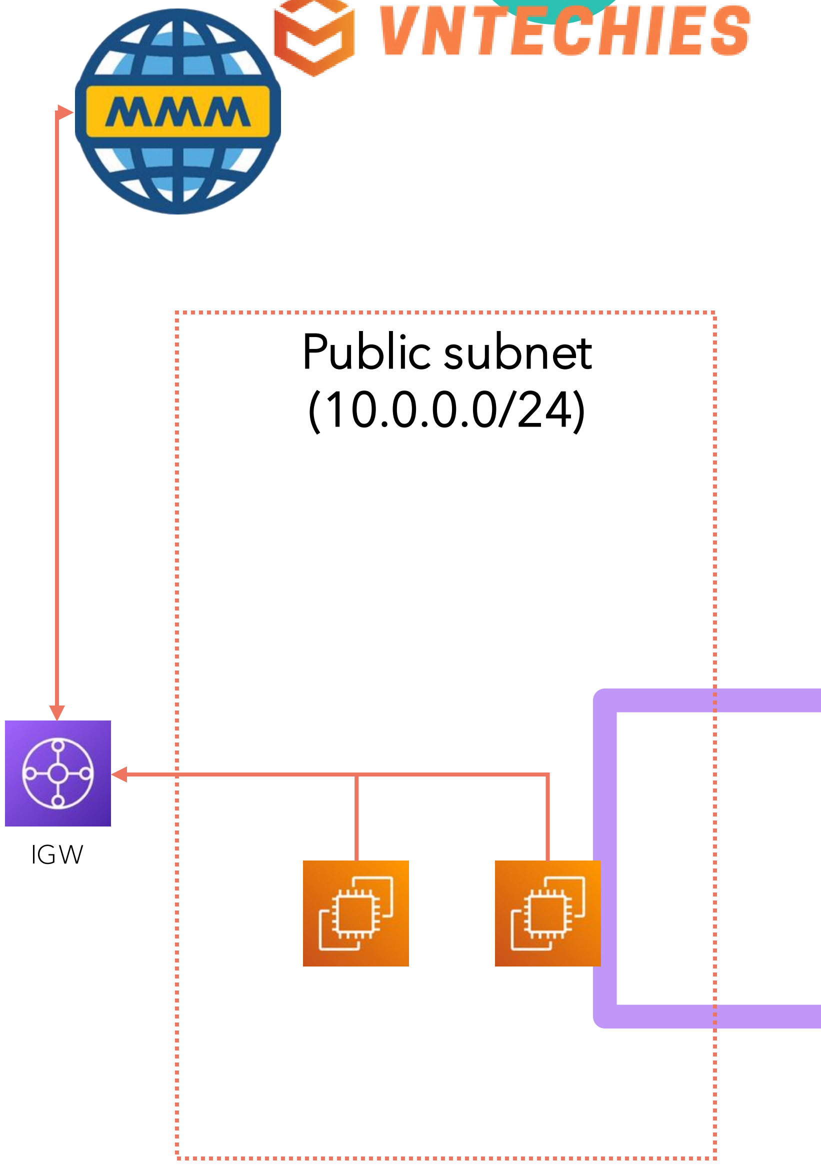
An **Internet Gateway (IGW)** is a component that connects your VPC to the **public internet**.

When attached to a VPC, it allows:

- **Inbound traffic** from the internet to reach *public subnets* (if the route table allows it), and
- **Outbound traffic** from instances to reach the internet.

Key Points:

- One VPC can have **only one Internet Gateway**.
- It must be **attached** to the VPC and **referenced in the route table** to work.
- Public subnets must route outbound traffic (0.0.0.0/0) through the IGW.

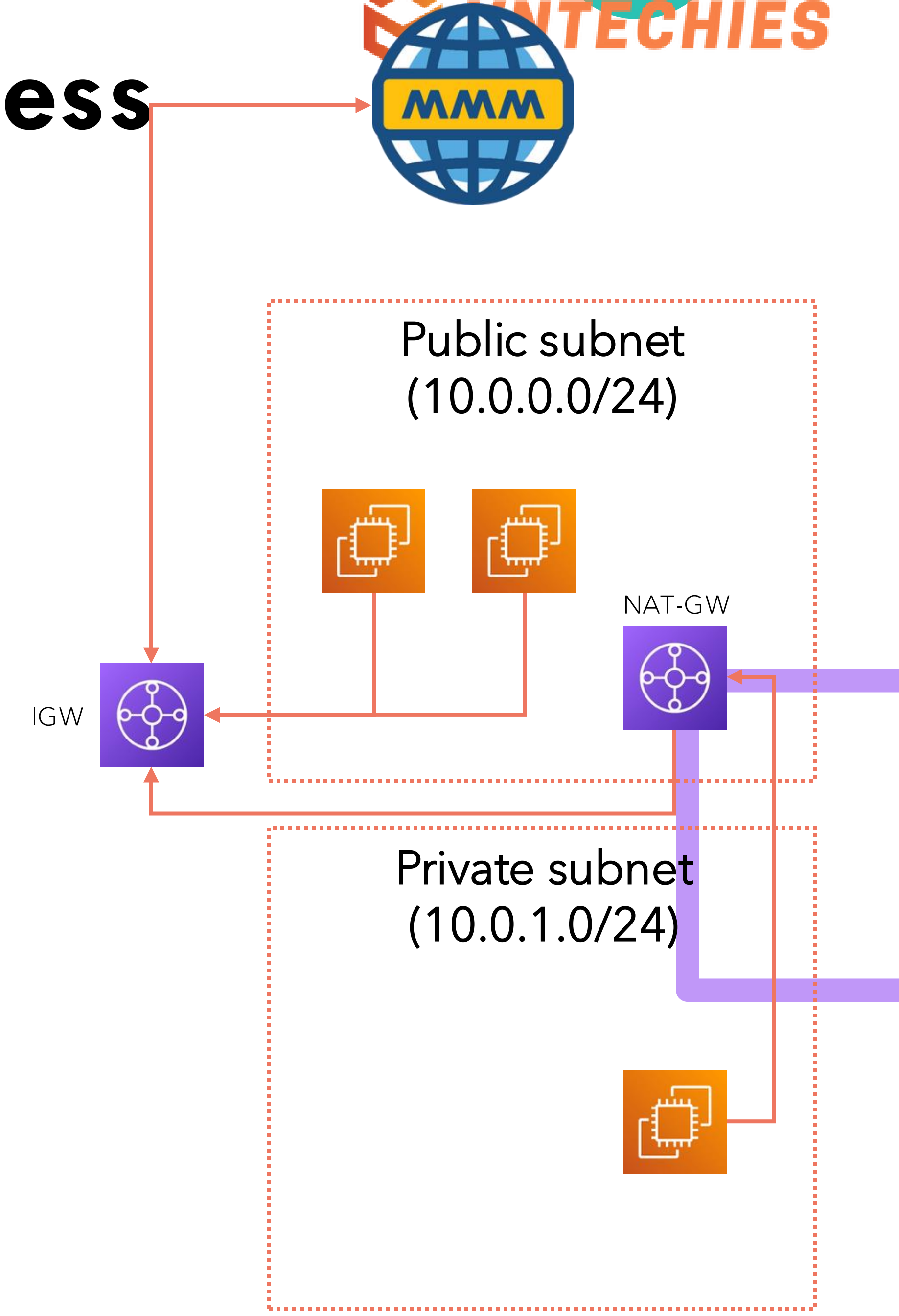


NAT Gateway (Network Address Translation Gateway)

A **NAT Gateway** allows resources in **private subnets** to access the internet *outbound* (e.g., to download updates or call APIs) **without being directly accessible from the internet**.

Key Behavior:

- It replaces **private IPs with a public IP** for outbound requests.
- It **does not allow inbound connections** initiated by the internet.
- It's fully managed and scales automatically.



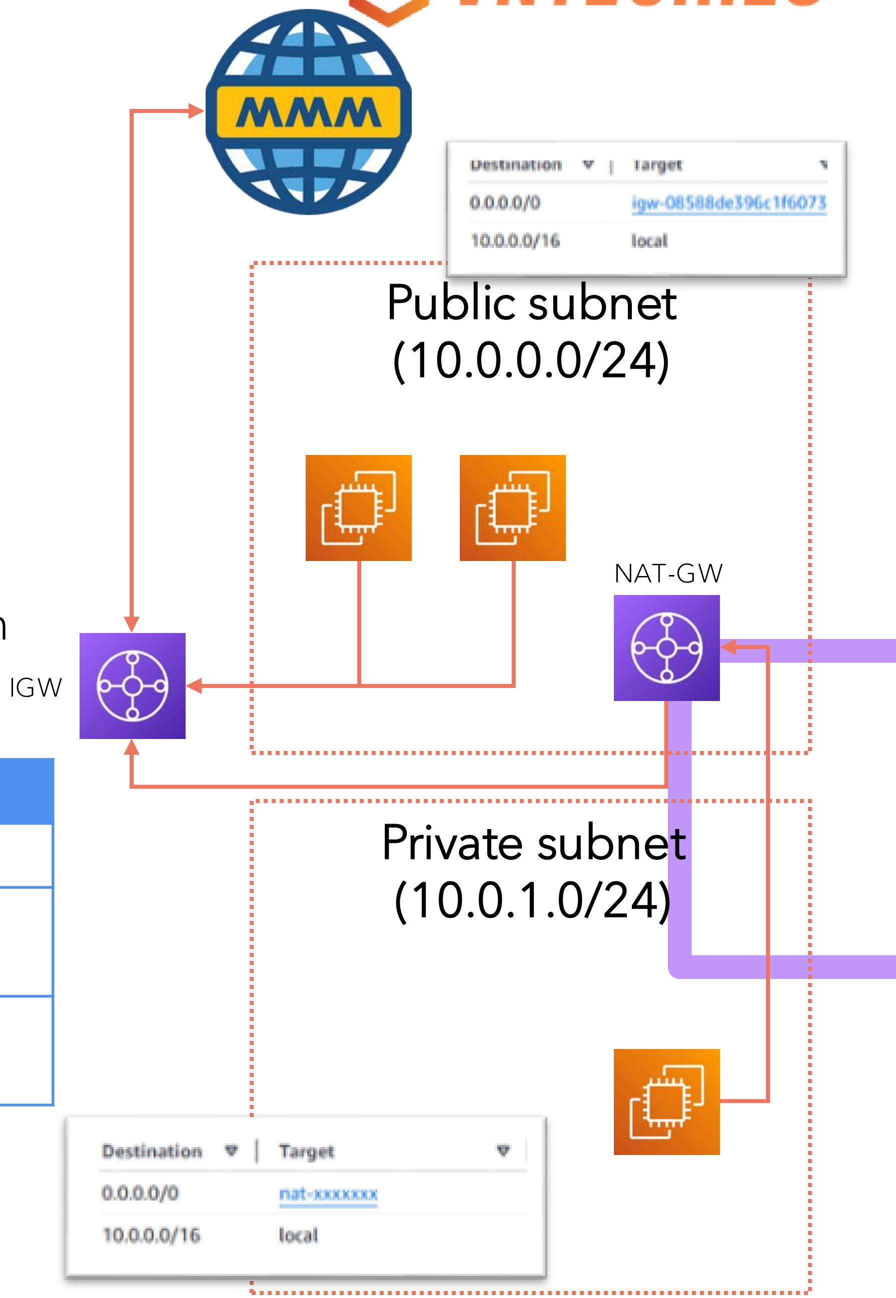
Route Table

Definition:

A **Route Table** contains a set of rules (routes) that determine **where network traffic is directed** within a VPC.

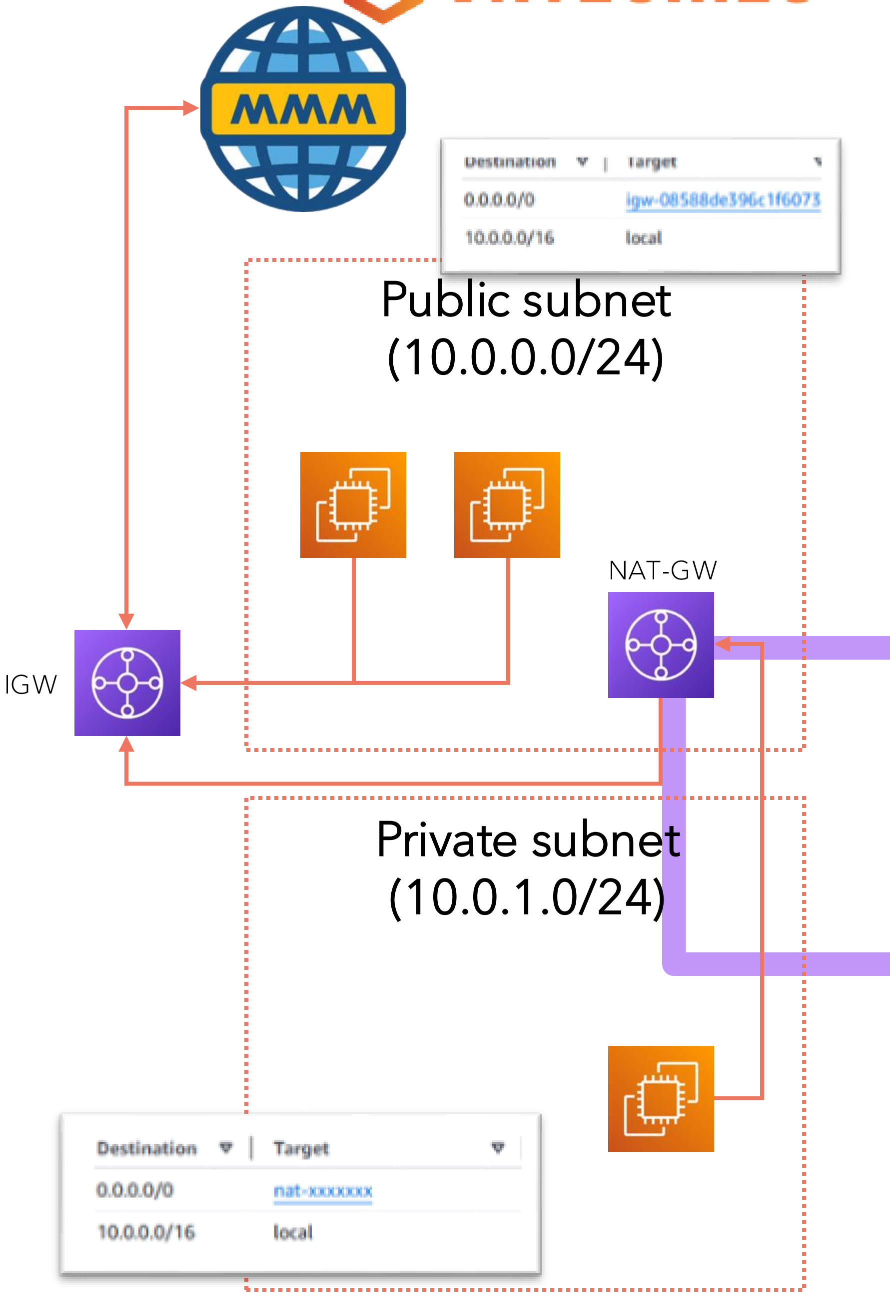
Each subnet is associated with one route table.
A route table decides whether a subnet is **public** or **private** based on where its default route (0.0.0.0/0) points.

Destination	Target	Purpose
10.0.0.0/16	local	Enables internal VPC communication
0.0.0.0/0	igw-12345	Sends all internet-bound traffic through the Internet Gateway (public subnet)
0.0.0.0/0	nat-67890	Sends internet-bound traffic from private subnets through the NAT Gateway



Summary

Component	Description	Example
Subnet	Logical network segment inside a VPC (Public or Private).	10.0.1.0/24 (public)
Internet Gateway (IGW)	Enables communication between VPC and internet.	igw-1234567890
NAT Gateway	Provides outbound internet access for private resources.	nat-0987654321
Route Table	Defines how traffic moves within or outside the VPC.	Public → IGW; Private → NATGW



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